Adult Income Classification: Baseline & Error Analysis Summary

1. Problem Framing

The objective is to classify individuals from the Adult Census dataset into two income brackets: earning `>50K` (Positive Class) and earning `<=50K` (Negative Class). The **business goal** is to maximize the precision among contacted users for a hypothetical marketing campaign, avoiding wasted outreach costs on individuals unlikely to afford the service or product.

2. Chosen Metric

The base rate for the positive class in our dataset is **23.93%**. Our primary evaluation metric for the project is the **F1-Score**, while maintaining a strict minimum threshold for **Precision**. We prioritize Precision over Recall because minimizing false positives directly reduces wasted marketing budget. However, optimizing exclusively for Precision can lead to zero Recall (identifying no one), so the F1-Score ensures we build a model that balances highly accurate targeting while still capturing a meaningful portion of the high-earner demographic.

3. Baseline Results

Two simple baselines were evaluated on a 20% hold-out test set (stratified by target) to establish a minimum performance floor:

- **Majority-Class Predictor:** Guesses `<=50K` for every individual.
  - *Accuracy:* 76.07% | *Precision:* 0.00% | *Recall:* 0.00% | *F1-Score:* 0.000
- **Rule-Based Predictor:** Predicts `>50K` if `education-num >= 13` (Bachelor's degree or higher).
  - *Accuracy:* 75.30% | *Precision:* 48.44% | *Recall:* 49.70% | *F1-Score:* 0.491

Interpretation: The single-feature rule easily outperforms the majority class predictor by actually attempting to find the positive class. However, its accuracy is slightly lower due to the introduction of False Positives. A real machine learning model must significantly improve upon an F1-Score of 0.49 and Precision of 48.4% to be considered useful for business outreach.

4. Error Analysis & Next Steps

We analyzed the False Positives (FP) and False Negatives (FN) generated by our education-based rule:

- **False Positives (Bachelors$+$, but earning $<=$50K):** Many are noticeably younger (avg. age $\sim$37) and work standard 40-hour weeks. They have degrees but lack the tenure and hours associated with top earners.

- **False Negatives (No degree, but earning $>$50K):** Many are older (avg. age ~44) and work longer hours (~45 hours/week). They are often highly experienced individuals in lucrative trades (`Craft-repair`) or hold management titles (`Exec-managerial`) and have high capital gains.

Concrete Next Steps for Modeling:

1. **Missing Values:** Impute or drop `NaN` values in features like `workclass` and `occupation` (originally marked as '?').
2. **Categorical Encoding:** Apply One-Hot Encoding to nominal variables (`occupation`, `marital-status`, `sex`) to capture non-linear trade roles.
3. **Skewed Numerics:** Apply log-transformations to highly skewed features like `capital-gain`.
4. **High Cardinality:** Group rare categories in `native-country` into an "Other" bin to prevent overfitting.
5. **Remove Redundancy:** Drop `fnlwgt` (as it is a survey weight, not a personal trait) and string `education` (keeping ordinal `education-num`).